Invariants, Monovariants, Coloring & Combinatorial Games

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Invariants, Monovariants, Coloring & Combinatorial Games Practice Exercises

1 The integers $1, 2, 3, \ldots, 10$ are written on a blackboard. At each step, two numbers a and b are erased and replaced by $a + b - 1$. After 9 steps, a single integer remains. What is that integer?

a. 45

b. 47

c. 46 ✓

d. 55

Notice the quantity $I = \sum(x_i - 1)$ is strictly invariant because $(a + b - 1) - 1 = (a - 1) + (b - 1)$. Equivalently, each of the 9 merges decreases the total sum by 1, so the final number is $\frac{10(11)}{2} - 9(1) = 46$.

2 The integers $1, 2, 3, \ldots, 11$ are written on a blackboard. At each step, two numbers a and b are erased and replaced by $a + b - 2$. After 10 steps, a single integer remains. What is that integer?

a. 44

b. 46 ✓

c. 48

d. 66

Notice the quantity $I = \sum(x_i - 2)$ is strictly invariant because $(a + b - 2) - 2 = (a - 2) + (b - 2)$. Equivalently, each of the 10 merges decreases the total sum by 2, so the final number is $\frac{11(12)}{2} - 10(2) = 46$.

3 The integers $1, 2, 3, \ldots, 12$ are written on a blackboard. At each step, two numbers a and b are erased and replaced by $a + b - 3$. After 11 steps, a single integer remains. What is that integer?

a. 45 ✓

b. 42

c. 48

d. 78

Notice the quantity $I = \sum(x_i - 3)$ is strictly invariant because $(a + b - 3) - 3 = (a - 3) + (b - 3)$. Equivalently, each of the 11 merges decreases the total sum by 3, so the final number is $\frac{12(13)}{2} - 11(3) = 45$.

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4 The integers $1, 2, 3, \ldots, 13$ are written on a blackboard. At each step, two numbers a and b are erased and replaced by $a + b - 4$. After 12 steps, a single integer remains. What is that integer?

a. 39 **b.** 47

c. 91 **d.** 43 ✓

Notice the quantity $I = \sum(x_i - 4)$ is strictly invariant because $(a + b - 4) - 4 = (a - 4) + (b - 4)$. Equivalently, each of the 12 merges decreases the total sum by 4, so the final number is $\frac{13(14)}{2} - 12(4) = 43$.

5 On a chalkboard are written the 8 fractions $\frac{1}{2}, \frac{1}{3}, \ldots, \frac{1}{9}$. Repeatedly, two numbers a and b are erased and replaced by $ab + a + b$ until one number remains. What is the exact final number?

a. 5 **b.** $\frac{8}{3}$

c. 4 ✓ **d.** $\frac{9}{2}$

Observe Simon's factorization $(ab + a + b) + 1 = (a + 1)(b + 1)$. Thus the product $\Pi = \prod(x_k + 1)$ is strictly invariant! Initially, $\Pi = \prod_{k=1}^{8}\left(1 + \frac{1}{1+k}\right) = \prod_{k=1}^{8}\frac{1+k+1}{1+k} = \frac{10}{2}$ by telescoping. Hence the final number x_{final} satisfies $x_{\text{final}} + 1 = \frac{10}{2} \Rightarrow x_{\text{final}} = 4$.

6 On a chalkboard are written the 9 fractions $\frac{1}{3}, \frac{1}{4}, \ldots, \frac{1}{11}$. Repeatedly, two numbers a and b are erased and replaced by $ab + a + b$ until one number remains. What is the exact final number?

a. 4 **b.** 3 ✓

c. $\frac{9}{4}$ **d.** $\frac{10}{3}$

Observe Simon's factorization $(ab + a + b) + 1 = (a + 1)(b + 1)$. Thus the product $\Pi = \prod(x_k + 1)$ is strictly invariant! Initially, $\Pi = \prod_{k=1}^{9}\left(1 + \frac{1}{2+k}\right) = \prod_{k=1}^{9}\frac{2+k+1}{2+k} = \frac{12}{3}$ by telescoping. Hence the final number x_{final} satisfies $x_{\text{final}} + 1 = \frac{12}{3} \Rightarrow x_{\text{final}} = 3$.

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7 On a chalkboard are written the 10 fractions $\frac{1}{4}, \frac{1}{5}, \ldots, \frac{1}{13}$. Repeatedly, two numbers a and b are erased and replaced by $ab + a + b$ until one number remains. What is the exact final number?

a. $\frac{5}{2}$ ✓

b. $\frac{7}{2}$

c. 2

d. $\frac{11}{4}$

Observe Simon's factorization $(ab + a + b) + 1 = (a + 1)(b + 1)$. Thus the product $\Pi = \prod(x_k + 1)$ is strictly invariant! Initially, $\Pi = \prod_{k=1}^{10}\left(1 + \frac{1}{3+k}\right) = \prod_{k=1}^{10}\frac{3+k+1}{3+k} = \frac{14}{4}$ by telescoping. Hence the final number x_{final} satisfies $x_{\text{final}} + 1 = \frac{14}{4} \Rightarrow x_{\text{final}} = \frac{5}{2}$.

8 On a chalkboard are written the 11 fractions $\frac{1}{5}, \frac{1}{6}, \ldots, \frac{1}{15}$. Repeatedly, two numbers a and b are erased and replaced by $ab + a + b$ until one number remains. What is the exact final number?

a. $\frac{16}{5}$

b. $\frac{11}{6}$

c. $\frac{12}{5}$

d. $\frac{11}{5}$ ✓

Observe Simon's factorization $(ab + a + b) + 1 = (a + 1)(b + 1)$. Thus the product $\Pi = \prod(x_k + 1)$ is strictly invariant! Initially, $\Pi = \prod_{k=1}^{11}\left(1 + \frac{1}{4+k}\right) = \prod_{k=1}^{11}\frac{4+k+1}{4+k} = \frac{16}{5}$ by telescoping. Hence the final number x_{final} satisfies $x_{\text{final}} + 1 = \frac{16}{5} \Rightarrow x_{\text{final}} = \frac{11}{5}$.

9 In normal-play Nim with three heaps of sizes $(3,\ 5,\ 7)$ (illustrated), a winning move exists on the third heap (size 7). How many stones must the first player **remove** from that heap of size 7 to leave a balanced Nim-sum 0 position?

a. 6

b. 3

c. 1 ✓

d. 2

Compute the binary Nim-sum $X = 3 \oplus 5 \oplus 7 = 1$. To restore Nim-sum 0 by playing on the heap of size 7, its new size must be $7 \oplus 1 = 6 < 7$. Therefore, the player must remove $7 - 6 = 1$ stones.

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10 In normal-play Nim with three heaps of sizes $(4,\ 7,\ 9)$ (illustrated), a winning move exists on the third heap (size 9). How many stones must the first player **remove** from that heap of size 9 to leave a balanced Nim-sum 0 position?

a. 3

b. 6 ✓

c. 10

d. 7

Compute the binary Nim-sum $X = 4 \oplus 7 \oplus 9 = 10$. To restore Nim-sum 0 by playing on the heap of size 9, its new size must be $9 \oplus 10 = 3 < 9$. Therefore, the player must remove $9 - 3 = 6$ stones.

11 In normal-play Nim with three heaps of sizes $(5,\ 8,\ 11)$ (illustrated), a winning move exists on the first heap (size 5). How many stones must the first player **remove** from that heap of size 5 to leave a balanced Nim-sum 0 position?

a. 2 ✓

b. 3

c. 6

d. -2

Compute the binary Nim-sum $X = 5 \oplus 8 \oplus 11 = 6$. To restore Nim-sum 0 by playing on the heap of size 5, its new size must be $5 \oplus 6 = 3 < 5$. Therefore, the player must remove $5 - 3 = 2$ stones.

12 In normal-play Nim with three heaps of sizes $(6,\ 9,\ 13)$ (illustrated), a winning move exists on the first heap (size 6). How many stones must the first player **remove** from that heap of size 6 to leave a balanced Nim-sum 0 position?

a. 4

b. 3

c. -2

d. 2 ✓

Compute the binary Nim-sum $X = 6 \oplus 9 \oplus 13 = 2$. To restore Nim-sum 0 by playing on the heap of size 6, its new size must be $6 \oplus 2 = 4 < 6$. Therefore, the player must remove $6 - 4 = 2$ stones.

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